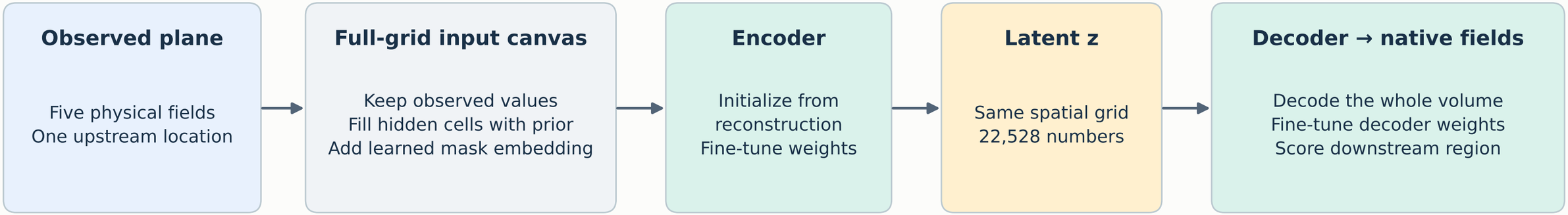
02 / Upstream → downstream

Question: can the pretrained spatial representation infer regions we do not observe?



Observed native plane: y=18 on 85604; y=17 on 85606; radial cells x ≥ 16.

Main task: infer the hidden downstream state

Input: upstream plane only
Loss: downstream truth versus predicted downstream fields

Optional clean replay — training only

Separate complete volume → same encoder/decoder → reconstruction
No hidden truth is supplied to the downstream input

With clean replay: downstream loss + 0.5 × clean reconstruction loss (the tested replay recipe).

Both tasks update the same weights. Replay is an extra training example, not an extra inference input.

Clean target-shot replay preserves a task; replaying source-shot examples is a separate choice for shot retention.